



THE BETTER POLICY PROJECT



AI-POWERED LEARNING

CBA WORKING PAPER 2023/19 · METHODOLOGY

Non-Tradeable Sticky Price Inflation

Developing Better Concepts for Monetary Policy Analysis

01

Concept & Rationale

NOEM foundations; sticky vs flexible, traded vs non-traded

02

Constructing the NTSPI

Tradability thresholds, basket composition & calculation

03

Policy Application

Endogenous credibility under FPAS Mark II

04

Historical Narrative

NTSPI in Armenia, 2007–2023



Decomposing the CPI for Policy

A decomposition that aligns with how monetary policy actually transmits

Headline, core, median and trimmed-mean measures aggregate prices from many distinct markets. The goal here is a split that aids both the execution and the communication of policy — one that maps onto the two most prominent aspects of the transmission mechanism.

ANCHOR 1

Expected policy-rate path

The decomposition should track the prices most relevant to the short-term interest-rate path — the slower-moving, expectations-driven components that policy can steer over time.

ANCHOR 2

Exchange-rate implications

Special attention goes to the currency channel. Internationally traded goods move with the exchange rate; separating them clarifies what policy is actually responding to.

EARLY-WARNING LOGIC

Why the split matters

Flexible prices adjust fast and warn early. If policy fails to respond to generalized excess demand, pressure spreads into stickier segments — wages and non-traded sticky prices — forcing a costlier adjustment later. 2021 is the cautionary case.



Foundations in Open-Economy Macro

Sticky and flexible prices defined economically — not statistically

- 1 **Dornbusch (1976) overshooting.** Fast-moving asset markets contrast with sluggish goods markets, producing exchange-rate overshooting. Traded goods — directly hit by the exchange rate — are flexible; non-traded items, mostly services, are sticky.
- 2 **NOEM microfoundations.** Obstfeld & Rogoff embedded these dynamics in New Open Economy Macroeconomics; the IMF's Global Economy Model (Laxton 2008) built on that path-breaking research.
- 3 **Economic, not statistical.** Core, median and trimmed-mean measures strip volatile items without an economic rationale. NOEM supplies one: exchange-rate-sensitive (flexible) versus less-sensitive (sticky) prices.

The key mapping

Traded goods → flexible prices
Exchange-rate driven; quick to adjust; idiosyncratic shocks.

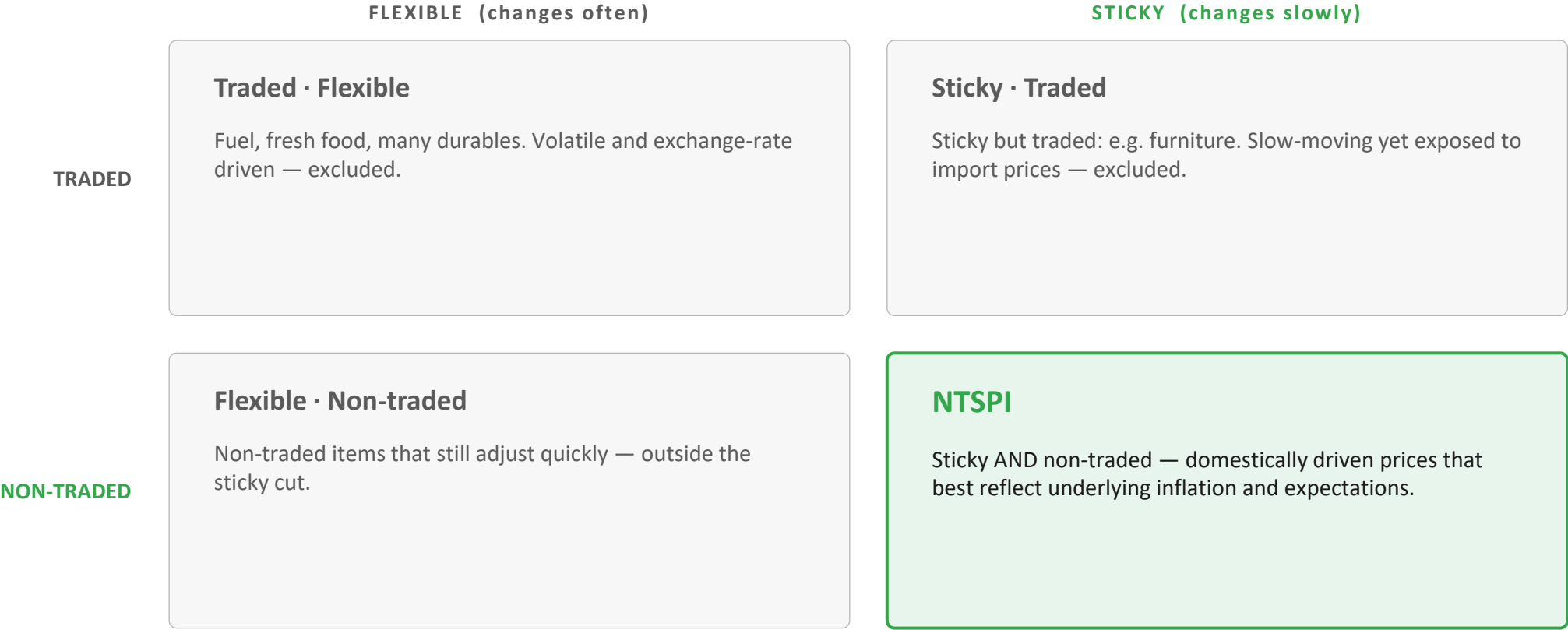
Non-traded goods → sticky prices
Most services; slow to adjust; reflect forward-looking expectations.



From CPI to a Better Underlying Measure

NTSPI keeps only the intersection of two independent cuts

The Atlanta Fed splits items by how often prices change; open-economy theory splits them by exposure to the exchange rate. NTSPI retains items that are both sticky and non-traded.





Looking Through Supply Shocks

Traded-goods volatility is often transitory and outside the reach of policy

- 1 Often transitory.** Traded prices move with exchange rates, global supply and demand, and supply-chain disruptions — frequently temporary or outside the purview of monetary policy, so they should not draw a period-by-period response.
- 2 The Covid case.** U.S. furniture rose 24% from April 2020 to April 2022 on stay-at-home demand and disrupted supply chains — then was among the first prices to decline once bottlenecks eased.
- 3 But watch the spillovers.** Policy should respond to where a shock leaks into persistent components. CPI, core and even sticky-price inflation all contain traded goods, adding volatility and complicating communication.

U.S. FURNITURE & BEDDING

+24%

cumulative price rise, Apr 2020 → Apr 2022 (in USD, NSA)

A durable, traded good buffeted by transitory supply shocks — exactly the kind of volatility NTSPI is designed to set aside.



Exchange-Rate Pass-Through & Dollarization

The currency channel, why it is tempered, and why non-traded prices can still be exposed

GEM DISTRIBUTION SECTOR

Pass-through is tempered

In the Global Economy Model (Laxton 2008), goods pass through a distribution sector, so import prices reflect both production and distribution costs.

A risk-premium shock's pass-through is slowed by inertias in retail-price and import-volume adjustment. Nominal and real rigidities reduce the CPI's sensitivity to the exchange rate — which must therefore move more in the short run.

FURNITURE, REVISITED

Much of it was the currency

Over the same window the U.S. dollar depreciated ~10% against the renminbi; China supplied ~42% of U.S. furniture imports (2019).

Priced in renminbi, furniture rose ~11% — against 24% in dollars. Roughly half the dollar increase was depreciation, not a fundamental supply–demand shift.

DOLLARIZATION CAVEAT

Non-traded ≠ always insulated

In highly dollarized economies such as Armenia, some non-traded prices — notably apartment rents — are set in U.S. dollars.

These become exchange-rate sensitive and more volatile. Capturing this matters: the aim is to eliminate excessive pass-through into non-traded prices, not to assume it away.



Building Block: Atlanta Fed Sticky-Price CPI

Items split by how frequently their prices change (Bryan & Meyer, 2010)

70%

sticky share of the CPI — prices change less often than every 4.3 months

30%

flexible share — mainly food and energy; prices adjust quickly

WHY STICKINESS IS INFORMATIVE

Prices that reset infrequently are reasonable candidates to embed forward-looking expectations about future inflation — so they may signal underlying inflation better than flexible prices (Bryan & Meyer, 2010).

WHAT THE STICKY BASKET CONTAINS

Service-based categories — personal care, medical services, education — and housing. The basket is built from the full CPI, classified item-by-item on the average frequency of price change.

THE LIMITATION NTSPI ADDRESSES

The sticky-price index still includes traded goods that move with the exchange rate. That confounds policy analysis and communication — motivating a further split into traded versus non-traded.



The Original Methodology

Classifying CPI items by how frequently their prices change

- 1 The evidence base.** Bils & Klenow (2004) measured the frequency of price change for the 350 detailed categories underlying the CPI. The median category resets its price every 4.3 months.
- 2 The separating rule.** A component is sticky if it changes price less often than every 4.3 months, and flexible otherwise — a natural cut-off at the median frequency.
- 3 The aggregate split.** On this rule ~70% of the headline CPI is sticky and ~30% flexible. About half the flexible share is food and energy; the rest is autos, apparel and lodging.

CLASSIFICATION RULE

$f_i > 4.3 \text{ months} \Rightarrow \text{sticky}$
 $f_i \leq 4.3 \text{ months} \Rightarrow \text{flexible}$

f_i = average months between price changes for component i (Bils & Klenow, 2004).

STICKY-PRICE INDEX

$$\pi_t(\text{sticky}) = \sum_i w_i \cdot \pi_{i,t}, \quad i \in S$$

The sticky set S is aggregated with CPI relative-importance weights w_i , renormalized to sum to one over S . NTSPI applies the same logic to the non-traded sticky subset.



Frequency of Price Adjustment

Each CPI component classified by its average months between price changes

FLEXIBLE · changes ≤ 4.3 months

Item	Mo.	Wt.
Motor fuel	0.7	3.2
Car & truck rental	1.2	0.1
Fresh fruits & vegetables	1.3	0.9
Fuel oil & other fuels	1.5	0.3
Gas (piped) & electricity	1.6	4.2
Meats, poultry, fish & eggs	1.9	1.9
Used cars & trucks	2.0	1.6
Leased cars & trucks	2.0	0.6
New vehicles	2.0	4.5
Women's & girls' apparel	2.3	1.5
Dairy & related products	2.6	0.9
Nonalcoholic beverages	2.7	1.0
Lodging away from home	3.1	2.5
Processed fruits & vegetables	3.2	0.3
Men's & boys' apparel	3.2	0.9
Cereals & bakery products	3.3	1.2
Footwear	3.4	0.7
Other food at home	3.6	2.0
Jewelry & watches	3.9	0.4
Motor vehicle parts & equipment	4.1	0.4
Tobacco & smoking products	4.2	0.8
Total flexible		29.8

STICKY · changes > 4.3 months

Item	Mo.	Wt.
Infants' & toddlers' apparel	5.3	0.2
Household furnishings & operations	5.3	4.8
Motor vehicle maintenance & repair	5.8	1.2
Motor vehicle insurance	5.9	2.0
Medical care commodities	6.2	1.6
Personal care products	6.7	0.7
Alcoholic beverages	7.3	1.1
Recreation	7.9	5.7
Miscellaneous personal goods	8.1	0.2
Communication	8.4	3.2
Public transportation	9.4	1.1
Tenants' & household insurance	10.1	0.3
Food away from home	10.7	6.5
Rent of primary residence	11.0	6.0
Owners' equiv. rent (4 regions)	11.0	24.4
Education	11.1	3.1
Medical care services	14.0	4.8
Water, sewer & trash	14.3	1.0
Motor vehicle fees	16.4	0.5
Personal care services	23.7	0.6
Miscellaneous personal services	25.9	1.1
Total sticky		70.1

Mo. = average months between price changes; Wt. = relative importance (% of CPI). Source: BLS; Bils & Klenow (2004); authors' calculations.



Sticky Prices in an Inflation Forecast

Why sticky prices carry information about where inflation is heading

EXPECTATIONS-AUGMENTED PHILLIPS CURVE

$$\pi_t = \beta \cdot E_t[\pi_{t+1}] + \kappa \cdot x_t$$

Inflation depends on expected inflation and on slack x_t . Two forces — expectations and the output/unemployment gap.

EMPIRICAL TEST (BRYAN & MEYER)

$$\pi_t = \alpha + \sum_k \beta_k \pi_{t-k}(\text{sticky}) + \sum_k \delta_k (u-u^*)_{t-k} + \varepsilon_t$$

Lagged sticky-price CPI proxies inflation expectations; the unemployment gap ($u-u^*$, CBO natural rate) measures slack. Up to 12 lags each.

FORECAST ACCURACY · RELATIVE RMSE

Horizon	Sticky	Core sticky	Core CPI
1 month	1.007	1.025	1.010
3 months	0.988	0.996	0.994
12 months	0.955	0.969	0.953
24 months	0.858	0.869	0.987

Values below 1.0 beat the headline-CPI benchmark. Sticky prices win at longer horizons — a 14% lower RMSE at 24 months — consistent with an embedded expectations component.

Out-of-sample 2000–2007, estimated 1983–1999. Source: Bryan & Meyer (2010), Tables 1–3.



What the Data Show: Two Forces

Sticky prices carry expectations; flexible prices carry the cycle

MONTHLY VARIANCE OF CPI AGGREGATES

Measure	Total sample	Since 1983
Flexible CPI	67.3	71.5
Core flexible CPI	19.9	15.8
Sticky CPI	10.6	1.7
Core sticky CPI	11.6	1.9
Sticky CPI ex-shelter	6.9	2.1

Flexible prices are ~8× more volatile than sticky (core). After 1983 the sticky variance collapsed (10.6 → 1.7) as expectations anchored — flexible volatility did not.

RESPONSE TO ECONOMIC SLACK

Regressing the 12-month CPI change on the unemployment gap ($u - u^*$, CBO natural rate):

Flexible CPI**slope < 0**

Strong negative relation — flexible prices respond to slack, the cyclical force.

Sticky CPI**slope ≈ 0 (+, n.s.)**

Slight, statistically insignificant positive slope — not slack-driven; it carries the expectations term instead.

Out-of-sample 2000–2007, estimated 1983–1999; up to 12 lags chosen by AIC; Newey–West HAC standard errors. Source: Bryan & Meyer (2010).



The NTSPI Synthesis

Atlanta Fed stickiness combined with BLS tradability

- 1 Two segmentations, combined.** NTSPI synthesises the Atlanta Fed sticky/flexible split with the BLS segmentation of traded versus non-traded goods — retaining only items that are both sticky and non-traded.
- 2 Driven by domestic forces.** The resulting measure is more sensitive to domestic factors: inflation expectations, labor-market conditions and aggregate demand — the variables monetary policy can influence.
- 3 First-round vs second-round.** Separating traded from non-traded sticky prices isolates the first-round effect of the exchange rate on traded goods from the second-round effect on non-traded goods.
- 4 A fit for FPAS Mark II.** It slots into FIT and FPAS practice, supporting communication of the central bank's price-stability commitment under significant uncertainty.

NTSPI

$= \textit{Sticky} \cap \textit{Non-traded}$

Strips away supply-shock and exchange-rate volatility, leaving the slower-moving, expectations-driven core of inflation that is most relevant for the policy-rate path.



Tradability Thresholds

Defining the traded / non-traded boundary draws on an established literature

DWYER (1992)

10% threshold

Tradability as the ease of trade; the threshold is set from economic, political and social factors. Imports or exports above 10% of output $\Rightarrow$ traded.

KNIGHT & JOHNSON (1997)

10% threshold

Extends Dwyer with exchange-rate volatility, domestic market size and degree of competition for a more nuanced boundary.

DIXON ET AL. (2004)

15% threshold

Statistics New Zealand input–output approach: industry shares of output, imports and exports; 15% judged the best cut-off.

JOHNSON (2017)

Classification basis

Goods subject to international competition vs domestic-only. A CPI excluding non-traded goods forecasts inflation better — the basis adopted here.

Across these studies the threshold sits at 10–15% of output traded; NTSPI adopts Johnson's classification approach, which improves inflation forecasts.



Composition of the Armenian NTSPI Basket

Entirely services and non-traded goods — services heavily dominate

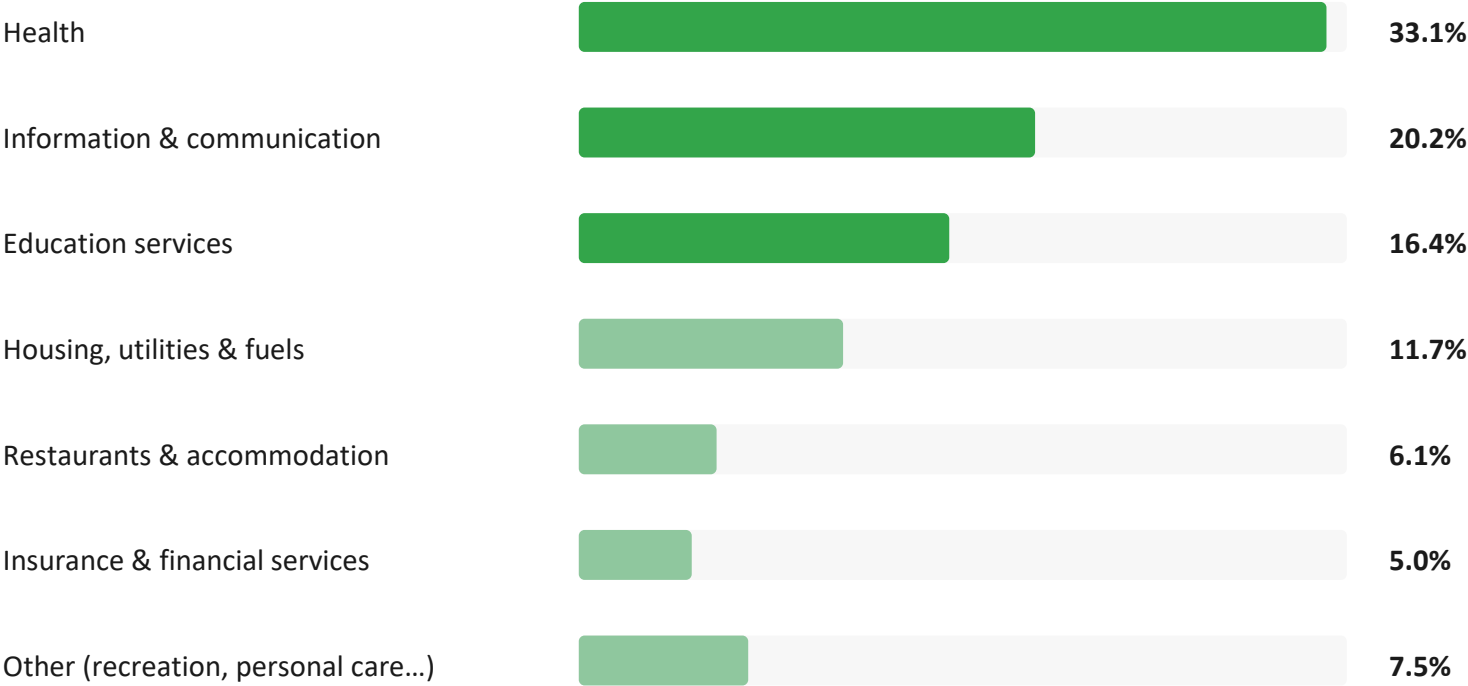
~15%

of the Armenian CPI basket is non-traded (2023)

12–22%

historical range of the non-traded share, 2006–2023

Health, information & communication, and education make up ~70% of the basket.

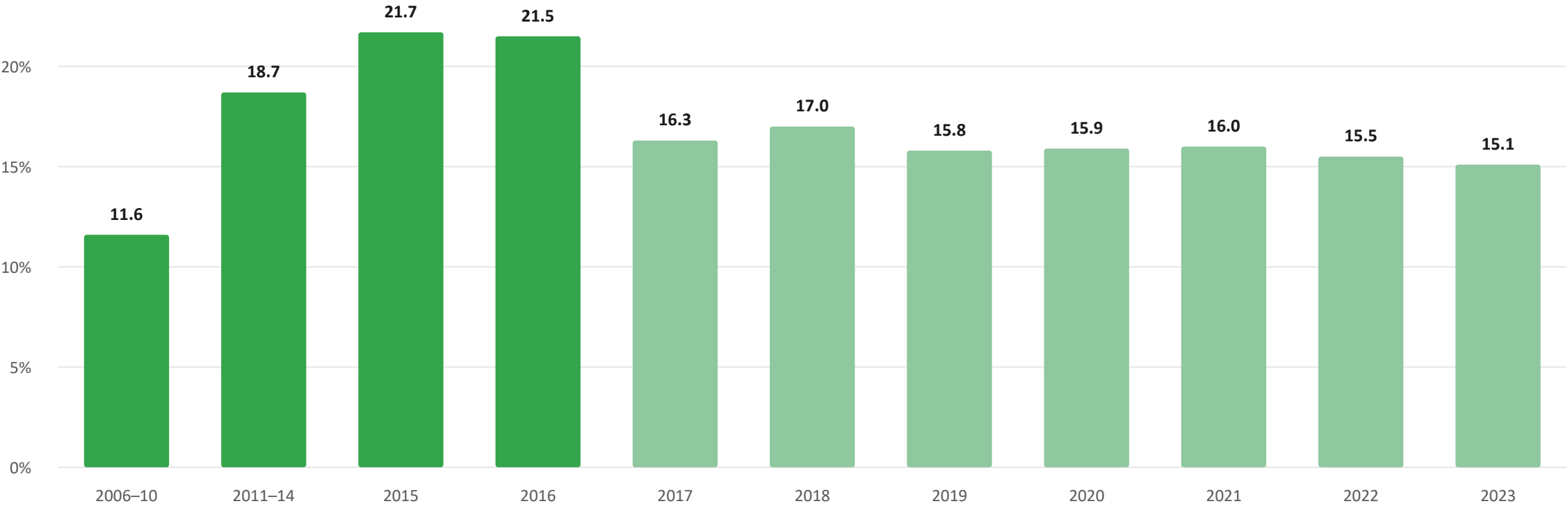


Share of NTSPI basket, by COICOP group. Source: author calculations, NSS Armenia.



The Non-Traded Share Over Time

Non-traded goods and services as a share of the Armenian CPI basket, 2006–2023



Peaks at ~22% during 2015–2016, then settles near 15–16% from 2019 onward. Source: author calculations, NSS Armenia.



Calculation Methodology

Same construction as the official Armenian CPI, on the non-traded sticky subset

01

Laspeyres, fixed weights

Built monthly with the Laspeyres formula and fixed base-period weights — identical to the official CPI method the NSS developed with the IMF.

02

Weights from the IHLCS

Basket weights come from household monetary-expenditure data in the Integrated Household Living Conditions Survey.

03

Two-year weight lag

The prior year is the base for indices and weights, but the survey lags two years — 2023 CPI weights are based on the 2021 IHLCS.

04

Non-traded sticky subset

The same machinery is applied only to items passing the tradability and stickiness criteria, producing the NTSPI index.

LASPEYRES INDEX

$$\text{NTSPI}_t = \sum_i w_i (p_{i,t} / p_{i,0})$$

Fixed base-period weights w_i over the non-traded sticky set; $p_{i,0}$ is the base-period price. Weights from the IHLCS, two-year lag.



Appendix · Composition of the Armenian NTSPI Basket

Item-level shares of the CPI and of the NTSPI basket

Item	COICOP Category	Share of CPI	Share of NTSPI
Secondary education (university) tuition	Education services	2.17%	14.27%
Surgical services	Health	1.56%	10.25%
Internet access provision services (monthly subscription)	Information & Communication	1.23%	8.11%
Bundled telecommunication services	Information & Communication	1.20%	7.86%
Actual rentals paid by tenants for secondary residences	Housing, water, electricity, gas & other fuels	1.13%	7.40%
Fast food and take away food services	Restaurants & accommodation	0.94%	6.15%
Actual rentals paid by tenants for main residence	Housing, water, electricity, gas & other fuels	0.61%	3.98%
Optometry services	Health	0.60%	3.97%
Cardiological services	Health	0.55%	3.64%
Annual bank card service fee	Insurance & financial services	0.51%	3.37%
Gynecology services	Health	0.35%	2.31%
Other Diagnosis Services (Covid)	Health	0.33%	2.15%
Wireless telephone services (prepaid)	Information & Communication	0.29%	1.90%
Funeral services	Personal care, social protection & misc.	0.28%	1.84%
Endocrinological services	Health	0.28%	1.82%
Dentistry	Health	0.25%	1.65%
Charges by banks and post offices	Insurance & financial services	0.25%	1.63%
General blood test	Health	0.23%	1.49%
X-Ray examination	Health	0.21%	1.36%
Dental prosthesis	Health	0.20%	1.32%
Wireless telephone services (postpaid)	Information & Communication	0.19%	1.28%
Urine analysis	Health	0.18%	1.16%



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Item-level shares of the CPI and of the NTSPI basket

Item	COICOP Category	Share of CPI	Share of NTSPI
Tomography	Health	0.15%	0.99%
Child care services	Personal care, social protection & misc.	0.15%	0.99%
Ultrasound examination	Health	0.15%	0.96%
Wired telephone services (monthly subscription)	Information & Communication	0.14%	0.92%
Women's hair cut	Personal care, social protection & misc.	0.14%	0.89%
Household textile and bed linen washing, dry cleaning	Furnishings & routine household maintenance	0.11%	0.72%
Preschool education tuition	Education services	0.09%	0.56%
Primary education	Education services	0.08%	0.54%
Secondary Vocational Education (College, Technical School)	Education services	0.08%	0.52%
Taxi transportation services	Transport	0.08%	0.52%
Elementary education	Education services	0.06%	0.40%
Automobile tire balancing services	Transport	0.05%	0.30%
Carpet washing, dry cleaning	Furnishings & routine household maintenance	0.04%	0.28%
Women's hairdressing and styling services	Personal care, social protection & misc.	0.04%	0.26%
Washing machines repair	Furnishings & routine household maintenance	0.04%	0.23%
Refrigerator repair	Furnishings & routine household maintenance	0.03%	0.21%
Women's hair dying services	Personal care, social protection & misc.	0.03%	0.21%
Men's haircut and styling services	Personal care, social protection & misc.	0.03%	0.20%
Car washing, polishing services	Transport	0.02%	0.16%
Repair of furniture, furnishings and floor coverings	Furnishings & routine household maintenance	0.02%	0.16%
Tiling services	Housing, water, electricity, gas & other fuels	0.02%	0.15%
Drywall and plaster services	Housing, water, electricity, gas & other fuels	0.02%	0.12%



Appendix · Composition of the Armenian NTSPI Basket

Item-level shares of the CPI and of the NTSPI basket

Item	COICOP Category	Share of CPI	Share of NTSPI
Legal services	Personal care, social protection & misc.	0.02%	0.10%
Shoe Repair	Clothing and footwear	0.01%	0.10%
Private foreign language courses	Education services	0.01%	0.09%
Driving courses, tests, licenses & technical-test fees	Transport	0.01%	0.08%
Dry Cleaning	Clothing and footwear	0.01%	0.08%
TV Repair	Furnishings & routine household maintenance	0.01%	0.07%
Wired telephone services (interregional fees)	Information & Communication	0.01%	0.05%
Color photo print services for passport	Personal care, social protection & misc.	0.01%	0.04%
Repair of jewellery, clocks and watches	Personal care, social protection & misc.	0.01%	0.04%
Tailoring	Clothing and footwear	0.00%	0.03%
Swimming pool-related services	Housing, water, electricity, gas & other fuels	0.00%	0.03%
Private computer courses	Education services	0.00%	0.03%
Repair of telephone or telefax equipment	Information & Communication	0.00%	0.02%
Cinema ticket	Recreation, sport, and culture	0.00%	0.02%
Zoo ticket	Recreation, sport, and culture	0.00%	0.02%
Services of painters	Housing, water, electricity, gas & other fuels	0.00%	0.01%
Repair of computers and computer equipment	Information & Communication	0.00%	0.01%
Television services (monthly subscription)	Information & Communication	0.00%	0.01%
Payments for copying documents	Information & Communication	0.00%	0.01%
Wired telephone services (per minute fees)	Information & Communication	0.00%	0.00%



Credibility as Endogenous

Under FPAS Mark II, policy actions themselves shape credibility

- 1 **Not fixed.** The workhorse ENDOCRED model treats credibility as endogenous: when policymakers and models ignore this, responses can be delayed or insufficiently aggressive.
- 2 **A reputation, earned.** Credibility means specifying a numerical long-term inflation objective and then achieving it on average over time — noise means inflation need not hit target every period.
- 3 **A stock.** It depends on accumulated performance. When expectations de-anchor it can be lost very quickly; regaining it is slow and gradual.
- 4 **Costly to lose.** If the public doubts the central bank, the policy rate must move far more aggressively — implying larger cumulative output and unemployment costs.

Why NTSPI fits

By design, NTSPI reflects the slower-moving, long-run drivers of inflation — above all the monetary-policy-relevant inflation expectations that are the key criterion for estimating and monitoring central-bank credibility.



Measuring Performance and Credibility

Two performance gauges combine into a credibility stock

NTSPI PERFORMANCE

$$(\text{NTSPI}_t - 10)^2 \div [(\text{NTSPI}_t - 10)^2 + (\text{NTSPI}_t - 2)^2]$$

Squared distance from a high-inflation regime (10%) versus the 2% target.
Closer to target → stronger performance.

DOLLARIZATION PERFORMANCE

$$(\text{D2}_t - 80)^2 \div [(\text{D2}_t - 80)^2 + (\text{D2}_t - 30)^2]$$

D2 = FX deposits & advances in residents' total. Distance from excessive (80%) versus optimal (30%) dollarization.

WEIGHTED PERFORMANCE INDICATOR

$$\text{CBPI}_t = \kappa \cdot \text{CBPI}(\text{NTSPI}) + (1 - \kappa) \cdot \text{CBPI}(\text{Dollar.})$$

A blend of the two gauges, with $\kappa = 0.7$ by default (slider range 0.5–1.0 for judgment).

CREDIBILITY AS A STOCK

$$\text{CRED}_t = 0.1 \cdot \text{CBPI}_t + 0.9 \cdot \text{CRED}_{t-1}$$

Credibility accumulates slowly from current performance — lost quickly, regained gradually.

Source: Papikyan, Avetisyan & Laxton (2023), Section IV.A.



NTSPI in Armenia, 2007–2023

A historical-narrative reading of CPI, Core and NTSPI

Because NTSPI reflects domestic demand and underlying inflation, it tells a different story than headline measures at key turning points. As of September 2023 NTSPI stood at 3.4% — above headline since April 2023 for the first time in the post-Covid era, even as headline dipped below target and into negative territory.

The policy challenge: reconcile receding headline inflation with still-elevated underlying inflation — ease too soon and risk re-acceleration; overstate it and risk a needless downturn.





Four Episodes

How NTSPI behaved — and what policy did — across four regimes

2006–2009

Double-digit growth

Capital inflows, ~50% appreciation and a real-estate boom (home prices +250%, 2003–07). NTSPI surged on demand and rising expectations — but stayed below CPI and Core. Target: 4%.

2009–2014

Slow growth, high risk premia

The GFC brought a sudden stop and debt-sustainability concerns. Flexible prices were volatile (gas, food); NTSPI stabilized near target. Sticky and flexible indices diverged after 2010.

2014–2020

Disinflation

An oil shock and ruble depreciation spilled over, destabilizing expectations. Policy tightened by ~12pp; an opportunistic disinflation from 2015 decelerated expectations and rebuilt credibility.

2020–2022

Post-war, post-pandemic

Covid plus the Nagorno-Karabakh war. The CBA raised rates in Dec 2020 — among the first globally — then by 650bp to a 10.75% peak (Dec 2022) as visitor- and remittance-driven demand pushed NTSPI up from early 2022.



Conclusion

NTSPI as one tool within a robust, narrative-driven framework

For a concept of sticky and flexible prices to be useful, it must align with how monetary policy influences the economy — and help answer three essential questions of good analysis:

1**State**

What is the current state of the economy?

2**Forces**

What are the underlying forces driving the economy and inflation?

3**Tools**

How should we set the policy rate and shape expectations of its path?

NTSPI is not a panacea or a single source of truth. It is one helpful tool that — inside a framework built on risk management, transparent communication and historical-narrative critical thinking — helps policymakers make and explain better-informed decisions.



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